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Photovoltaic thermal collector adaptor

Abstract

An adaptor for adapting a solar heater to a thermal storage device, the adaptor including a first port disposed on a first surface, a second port disposed on a second surface, a third port disposed on the second surface, and a fourth port disposed on the first surface, wherein a fluid flow is configured to be directed from the solar heater through the first port before exiting through the second port into the thermal storage device, the fluid flow is configured to continue to be directed from the thermal storage device through the third port before entering through the third port and continuing to exit through the fourth port to return to the solar heater.

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Background/Summary

PRIORITY CLAIM AND RELATED APPLICATIONS (1) This continuation-in-part application claims the benefit of priority from non-provisional application U.S. Ser. No. 18/119,344 filed Mar. 9, 2023. Said application is incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

1. The Field of the Invention
- (1) The present invention relates to a photovoltaic thermal (PVT) collector adaptor. More specifically, the present invention is directed to a PVT collector adaptor for functionally connecting a PVT collector to a thermal battery or for functionally connecting a PVT collector to a thermal battery and a heat source.
2. Background Art
- (2) Although solar heating technologies have existed for many years, drawbacks to conventional solar heating systems have hampered wide scale adoption of the technology. For example, due to their inefficiencies in interconnections of solar heaters to thermal batteries, conventional solar heating systems require custom connecting solutions which indirectly require significant levels of knowledge and understanding on the part of the end users or even professional installers of solar heating systems in order to assemble a solar heating system, especially a solar heating system involving electricity and thermal storage devices. This necessitates the determination of appropriate

interconnections required of solar heaters and photovoltaic cells and thermal batteries and additionally one or more power sources required to make the solar heaters and thermal batteries a functional system. Additionally, the use of complex and custom designed components in many conventional solutions, requires complicated interconnections between fluid transmitting parts, increasing the initial manufacturing and installation costs, as well as long term maintenance costs if components for interconnections have not been sourced properly. Another drawback to conventional systems is that consumers or even installers who are required to source photovoltaic thermal (PVT) collector systems for consumers are often deterred by the impact of the unsightly and disorganized assortment of fluid conductors, electrical harnesses and other cooperating devices have on the aesthetics of their properties on which the PVT collector systems are installed. Further, mistakes caused by poor training in the installations of such systems and the complicated interconnections between components of such systems can mean resulting systems which do not perform to their full potential and resulting systems which do not operate altogether.

(3) It would be advantageous to provide an adaptor suitable for connecting a photovoltaic thermal collector to a thermal battery to form a functional solar thermal heating system or an adaptor suitable for connecting a solar heater and a photovoltaic collector to a thermal battery and an electric battery to form a solar thermal heating system that can also receive thermal energy from one or more additional heat sources.

SUMMARY OF THE INVENTION

(4) In accordance with the present invention, there is provided an adaptor for adapting a solar heater to a thermal storage device, the adaptor including: (a) a first port disposed on a first surface; (b) a second port disposed on a second surface; (c) a third port disposed on the second surface; and (d) a fourth port disposed on the first surface,

wherein a fluid flow is configured to be directed from the solar heater through the first port before exiting through the second port into the thermal storage device, the fluid flow is configured to continue to be directed from the thermal storage device through the third port before entering through the third port and continuing to exit through the fourth port to return to the solar heater.

(5) In one embodiment, the adaptor further includes a pump configured to direct the fluid flow from the solar heater to the thermal storage device and from the thermal storage device to the solar heater. In one embodiment, the adaptor further includes a controller configured to control the pump. In one embodiment, an inlet of the pump is configured to be disposed at a level substantially the same as the level at which the third port is disposed and the third port is disposed at a level lower than the second port. In one embodiment, at least one of the second port and the third port is configured for a quick connection or disconnection with the thermal storage device. In one embodiment, at least one of the second port and the third port includes a groove disposed about a periphery of the at least one of the second port and the third port and a seal disposed within the groove. In one embodiment, the seal includes an O-ring. In one embodiment, the positioning of at least one of the second port and the third port relative to the second surface is configured to be adjustable. In one embodiment, the adaptor further includes an electrical receptacle for receiving an electrical connection from a solar electric generator. In one embodiment, the adaptor further includes an electric storage device configured to receive and store electrical charge from the solar electric generator. In one embodiment, the adaptor further includes an inverter configured to allow backfeeding of electricity from the electric storage device to an electrical grid through the inverter. In one embodiment, the adaptor further includes a hook configured for interfacing with a latch disposed on an enclosure of the thermal storage device such that a portion of the weight of the adaptor is supported on the enclosure while the second port and the third port are fluidly coupled with the thermal storage device. In one embodiment, the adaptor further includes a fifth port fluidly connected to the first port, wherein the fifth port is configured to be a bleed port. In one embodiment, the adaptor further includes a sixth port fluidly connected to the fourth port, wherein the fifth port and the sixth port are configured to be fluidly connected to a heat source. In one

embodiment, the positioning of at least one of the second port and the third port relative to the second surface is configured to be adjustable. In one embodiment, at least a portion of a fluid conductor connecting the first port and the second port is a flexible fluid conductor. In one embodiment, at least a portion of a fluid conductor connecting the third port and the fourth port is a flexible fluid conductor. In one embodiment, the first surface is disposed in a plane different from the second surface.

(6) An object of the present invention is to provide an adaptor suitable to functionally connect a photovoltaic thermal collector and a thermal battery.

(7) Another object of the present invention is to provide an adaptor that is uncomplicated and suitable to functionally connect a photovoltaic thermal collector and a thermal battery.

(8) Another object of the present invention is to provide an adaptor suitable to not only functionally connect a solar heater to a thermal battery but also functionally connect a solar electric generator to an electric battery.

(9) Whereas there may be many embodiments of the present invention, each embodiment may meet one or more of the foregoing recited objects in any combination. It is not intended that each embodiment will necessarily meet each objective. Thus, having broadly outlined the more important features of the present invention in order that the detailed description thereof may be better understood, and that the present contribution to the art may be better appreciated, there are, of course, additional features of the present invention that will be described herein and will form a part of the subject matter of this specification.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order that the manner in which the above-recited and other advantages and objects of the invention are obtained, a more particular description of the invention briefly described above will be rendered by reference to specific embodiments thereof which are illustrated in the appended drawings. Understanding that these drawings depict only typical embodiments of the invention and are not therefore to be considered to be limiting of its scope, the invention will be described and explained with additional specificity and detail through the use of the accompanying drawings in which:

(2) FIG. 1 is a diagram depicting one embodiment of a present adaptor configured to connect a photovoltaic thermal collector to a thermal battery.

(3) FIG. 2 is a diagram depicting one embodiment of a present adaptor being used to connect a photovoltaic thermal collector to a thermal battery.

(4) FIG. 3 is a cross-sectional view of the third port depicting one embodiment of an adjustment mechanism which allows the position of the third port to be adjustable relative to the surface upon which the third port is mounted.

(5) FIG. 4 is a diagram depicting one embodiment of a present adaptor having been used to connect a photovoltaic thermal collector to a thermal battery.

(6) FIG. 5 is a diagram depicting one embodiment of a present adaptor having been used to connect a photovoltaic thermal collector to a thermal battery.

(7) FIG. 6 is a diagram depicting one embodiment of a present adaptor in which a pump useful for circulating fluids through the present adaptor is configured to be easily accessible for replacement.

PARTS LIST

(8) 2—adaptor 4—port 6—port 8—port 10—port 12—pump 14—conductor 16—conductor 18—thermal storage device or battery 20—port 22—port 24—working fluid 26—vent 28—photovoltaic thermal collector 30—fastener 32—conductor 33—conductor 34—connector 36—connector 38—surface 40—surface 42—seal, e.g., O-ring seal 44—electrical connector 46—electrical receptacle

48—circuit breaker **50**—controller **52**—electricity storage device, e.g., electric battery **54**—electric grid **56**—inverter **58**—power supply **60**—hook **62**—latch **64**—heater, e.g., heat pump **66**—heater, e.g., resistive heater **68**—enclosure **70**—slot **72**—solar charge controller **74**—auxiliary input port **76**—port for receiving heated flow from heat source **78**—port for allowing spent flow to heat source **80**—input power port to heater **82**—stand-off **84**—output power port **86**—heat exchanger **88**—port **90**—port **92**—groove

PARTICULAR ADVANTAGES OF THE INVENTION

(9) The present adaptor allows a suitable photovoltaic thermal collector to be functionally connected to a thermal battery without requiring modifications to be made to any part of the photovoltaic thermal collector, thermal battery and electric battery, if one is utilized. The present adaptor simplifies connections between a photovoltaic thermal collector and a thermal battery, removing or reducing the need for a trained installer for installation of a photovoltaic thermal (PVT) system. In one embodiment, a present adaptor further simplifies connections between a photovoltaic thermal collector with an electrical grid, allowing surplus electricity to be supplied to the electrical grid from the photovoltaic thermal collector.

DETAILED DESCRIPTION OF A PREFERRED EMBODIMENT

(10) The term “about” is used herein to mean approximately, roughly, around, or in the region of. When the term “about” is used in conjunction with a numerical range, it modifies that range by extending the boundaries above and below the numerical values set forth. In general, the term “about” is used herein to modify a numerical value above and below the stated value by a variance of 20 percent up or down (higher or lower).

(11) FIG. 1 is a diagram depicting one embodiment of a present adaptor **2** configured to connect a photovoltaic thermal (PVT) collector or a combined solar electric generator and a solar heater to a thermal battery. The adaptor **2** is useful for adapting a photovoltaic thermal collector **28** for heating a working fluid stored in a thermal storage device **18**. The adaptor **2** is essentially a body including interfaces such as a first port **4**, a second port **6** fluidly connected to the first port **4**, a third port **8** and a fourth port **10** fluidly connected to the third port **8** and a pump **12** configured for receiving a fluid through the third port **8** and pulling the fluid through the third port **8** from a thermal storage device **18** to the fourth port **10**. The first port **4** and the fourth port **10** are configured to be supplied on a first surface **38** and the second port **6** and third port **8** are configured to be supplied on a second surface **40** of the body. The first surface **38** is preferably a surface disposed in a different plane from the second surface **40** as the first surface **38**, upon the installation of the adaptor, will no longer be accessible. With a present adaptor **2**, a technician or installer of the PVT heating system needs only to ensure that ports on one surface are connected to fluid conductors of the solar heater and ports on the other surface are inserted into ports disposed on one side of the thermal battery, removing any guesswork on the part of the technician or installer and installation mistakes that can cause a drop in efficiency of the operations of the heating system and also mistakes that can also prevent optimal operations of the heating system, can be avoided. The body can be a block having a plurality of surfaces including the first surface **38** and second surface **40**. No working fluid is shown disposed in the thermal battery **18** of FIG. 1. In the embodiment shown, the adaptor **2** further includes an electrical receptacle **46** for receiving an electrical connection via an electrical connector **44** from the solar electric generator. Therefore, in addition to the fluid connections via ports **4**, **6**, **8** and **10**, there is provided an electrical connection which facilitates the connection of the solar electric generator with the necessary equipment in the adaptor that it interfaces with. This further eliminates any guesswork on the part of the installer. In the embodiment shown, the electrical receptacle **46** is electrically connected to a solar charge controller **72** which handles the charging of an electric storage device, e.g., a battery **52**. This battery **52** is useful for powering a controller **50** which controls and supplies power to the pump **12** that drives a working fluid **24** through the conductor that connects the third port **8** and the fourth port **10** to the solar heater, to be heated in the solar heater, before returning through the conductor that connects the first port **4** and

the second port **6** into the thermal battery **18**. Without an external power source, e.g., power from an electrical grid **54**, the PVT can function, provided that there is sufficient solar power received by the solar electric generator and solar heater. However, with access to an electrical grid **54** protected by circuit breaker **48**, the PVT can be configured to not only transfer sufficient solar power for its own use, but also backfeed surplus electricity generated by the solar electric generator to the electrical grid **54** once the battery **52** it is configured to charge has been fully charged. An inverter **56** is interposed between the solar charge controller **72** and a power supply **58** which alternatively or additionally provides power to the controller **50** and the pump **12**. An additional output power port **84** may also be provided to power heater **64**, e.g., a heat pump, by connecting port **84** to port **80** of heater **64** to power the compressor of the heat pump. With access to the electrical grid **54**, the heating system can be operational even if the battery **52** has been insufficiently charged from which to draw power, to power the pump **12** as the power supply **58** which receives its input power from the inverter **56** is capable of supplying not only power to the pump **12** for circulating the working fluid **24** but also any devices operationally connected to it. In one embodiment, the inlet of the pump **12** is configured to be disposed at a level substantially the same, e.g., within about 1-2 inches, as the level at which the third port **8** is disposed or at a level as far below as possible from the second port **6**, to afford the pump **12** sufficient or maximum head during operation and as much head as possible for self-priming during the first start-up of the pump **12**. It shall be noted that the third port **8** is disposed at a level lower than the second port **6**.

(12) In the embodiment shown, each of the second port **6** and the third port **8**, is configured for a quick connection or disconnection with the thermal storage device **18**. A quick connection or disconnection between two mating ports allows the two mating ports to be coupled to form a leak-free fluid connection or the two mated ports to be disconnected without requiring a significant amount of effort, reducing the amount of effort, time and expense for assembling/installing or dismantling a heating system utilizing such a mechanism. In order to achieve liquid-tight connections while allowing easy insertion of one port in the other, each of the second port **6** and the third port **8** is equipped with a seal about its periphery. In order to retain a seal on each, a groove **92** is disposed about a periphery of each such that a seal **42**, e.g., an O-ring, can be disposed within the groove and retained therein as shown in FIG. 3.

(13) FIG. 2 is a diagram depicting one embodiment of a present adaptor **2** being used to connect a PVT collector to a thermal battery. As shown in FIG. 2, the alignment of the second port **6** with respect to receiving port **22** and the third port **8** with respect to receiving port **20** are further facilitated using a hook **60** configured for interfacing with a latch **62** disposed on an enclosure **68** of the thermal storage device **18** such that a portion of the weight of the adaptor **2** is supported on the enclosure **68** while the second port **6** and the third port **8** are fluidly coupled with the thermal storage device **18**. A receiving port **20**, **22** is essentially a cylindrical tube configured to slidably receive a seal-equipped port, e.g., port **8**, **6**, respectively. Each of the hook and the latch is essentially a part having a cross-sectional profile shown but a part which extends orthogonally from the cross-sectional profile. An installer of the heating system can first focus on hanging the adaptor **2** on the enclosure **68** at roughly the level the adaptor **2** is to be coupled to the thermal storage device **18**. As the ports, once connected, interface with a working fluid, e.g., a glycol mixture, care must be taken to ensure that the ports are connected snugly to prevent leakages. The hook-and-latch mechanism affords the installer time to adjust the positioning of at least one of the ports **6**, **8** properly as the installer does not need to be concerned with lifting and supporting the entire weight of the adaptor, due to the use of the hook **60** and latch **62** to alleviate the challenge of installing the adaptor by aligning ports **6**, **8** with receiving ports **22**, **20**, respectively while supporting the full weight of the adaptor. A stand-off **82** is provided substantially on an opposing end of the surface **38** from the hook **60** to ensure that, once seated as shown elsewhere herein, the adaptor **2** no longer has the tendency to rotate to cause dislodgement of the second port **6** from port **22** and any potential rattling that can occur due to contacts of the adaptor **2** and the enclosure **68** caused by

wind and vibrations from equipment, e.g., pump **12**, can be minimized. The adaptor **2** is preferably further secured at the stand-off **82** to the enclosure **68** using, e.g., one or more fasteners. In one embodiment, the second port **6** is fixedly positioned on surface **38** while the third port **8** is configured to be adjustable. Despite efforts in ensuring precision manufacturing practices have been adhered to, mechanical parts are often still being made with a wide range of tolerances. Heating system parts are no exception. Therefore, it is imperative that for the adaptor **2** to be mateable by way of the thermal battery ports **20**, **22**, the position of at least one of the second port **6** and third port **8** is adjustable. FIG. **3** is a cross-sectional view of the third port **8** depicting one embodiment of an adjustment mechanism which allows the position of the third port **8** to be adjustable relative to the surface **38** upon which the third port **8** is mounted. Here, the third port **8** includes a flange having a plurality of slots **70**, one disposed on each side of the third port **8**. These slots **70** allow the port **8** to be slid up or down along surface **38** of the adaptor **2** before the fasteners **30** are tightened to secure the third port **8** in place. In another embodiment, the positioning of the second port **6** is made adjustable instead. In yet another embodiment, the positioning of each of the second port **6** and the third port **8** relative to surface **38** is made adjustable. In one embodiment, at least a portion of a fluid conductor **16** connecting the first port **4** and the second port **6** is a flexible fluid conductor which makes positional adjustment of the second port **6** easier. In one embodiment, at least a portion of a fluid conductor **14** connecting the third port **8** and the fourth port **10** is a flexible fluid conductor which makes positional adjustment of the fourth port **10** easier.

(14) FIG. **4** is a diagram depicting one embodiment of a present adaptor **2** having been used to connect a photovoltaic thermal collector to a thermal battery **18**. Heated flow of the solar heater is received by the thermal battery **18** via fluid conductor **33** and connector **36** that is mated with port **4** while spent flow from the thermal battery **18** is sent through connector **34** and fluid conductor **32** back to the solar heater. It shall be noted that the thermal battery **18** is now filled with a working fluid **24**, e.g., a glycol mixture, the interior of the thermal battery **18** is vented using a vent **26**. In one embodiment, the adaptor **2** further includes a fifth port **74** fluidly connected to the third port **8**, the fifth port **74** is configured to be a bleed port to allow the pump **12** to be self-primed at the first start-up upon installation with the working fluid, e.g., a glycol mixture used to fill the thermal battery **18**. In one embodiment, once the thermal battery **18** has been deemed operational, the fifth port **74** is capped to prevent evaporation of the contents of the thermal battery **18** through it. The heating system is transported with the thermal battery **18** void of a working fluid **24** to save on transportation costs and other expenses due to the transportation of a liquid which can be constituted onsite. Once the adaptor **2** has been coupled with the thermal battery **18** and the PVT, the pump **12** can be turned on to circulate a working fluid **24** to transport heated working fluid from the solar heater to the thermal battery **18**. In one embodiment, this process continues until the controller **50** determines that the thermal battery **18** has been sufficiently thermally charged, at which point the pump **12** is then turned off. Once charged, the thermal battery **18** becomes a heat source to heat a cold water supply requested through heat exchanger, e.g., coil **86**. The adaptor **2** can additionally serve as an electrical and/or additionally fluid interface for an additional heater, e.g., a resistive heater **66** and a heat source, e.g., a heat pump **64** circulating the working fluid **24** by emptying heated working fluid through auxiliary port **76** and drawing spent working fluid **24** through port **78**. If necessary, both ports **74**, **76** can be utilized to add an additional heat source, e.g., another heat pump to increase the rate at which the thermal battery **18** can be charged.

(15) FIG. **5** is a diagram depicting one embodiment of a present adaptor **2** having been used to connect a photovoltaic thermal power generator to a thermal battery **18**. Here, the adaptor **2** is suitable for use in locales where no electrical grid **54** or municipal electric power is available. Devices required for accessing an electrical grid is absent as electric energy required to power the pump **12**, is drawn directly from the solar electric generator. If the battery **52** has become fully charged and the thermal battery **18** has become charged to the desired level, no part of the PVT **28** is able to continue to harvest and store energy in the forms of electrical energy and thermal energy.

It can be seen in all the embodiments disclosed herein that the adaptor 2 reduces what would otherwise be a heating system largely unfamiliar to installation technicians of conventional heating systems of other fuels, e.g., natural gas and propane, etc., to mating parts and action steps that are intuitive to be carried out to set up a PVT to a thermal battery, reducing the potential for mistakes and increasing the efficiency in setting up the PVT heating system.

(16) FIG. 6 is a diagram depicting one embodiment of a present adaptor in which a pump 12 useful for circulating fluids through the present adaptor 2 is configured to be easily accessible, e.g., for replacement. Here, the pump 12 is shown connected via ports 88 and 90 of the adaptor 2 to the adaptor 2 where the pump 12 is disposed exterior of the adaptor 2. Therefore, it shall be understood that the pump 12 may be interposed in conductor 14 either exterior of the adaptor 2, or interior of the adaptor 2, as shown elsewhere herein. Further, the pump 12 need not be treated as an integral part of the adaptor 2 as the pump 12 may be replaced. Such replacement may be easier to perform when the pump 2 is disposed on the exterior of the adaptor 2. A controller 50 is shown functionally connected to the pump 12 as well, on the exterior of the adaptor 2. As shown throughout herein, a connection with a controller 50 can imply, in addition to a signal level connection, e.g., a connection useful for supplying signals to the pump 12 for controlling the actions of the pump 12, the same connection can be used to imply a power supply connection as well, where sufficient power is configured to be supplied to the pump 12 for its operation.

(17) The detailed description refers to the accompanying drawings that show, by way of illustration, specific aspects and embodiments in which the present disclosed embodiments may be practiced. These embodiments are described in sufficient detail to enable those skilled in the art to practice aspects of the present invention. Other embodiments may be utilized, and changes may be made without departing from the scope of the disclosed embodiments. The various embodiments can be combined with one or more other embodiments to form new embodiments. The detailed description is, therefore, not to be taken in a limiting sense, and the scope of the present invention is defined only by the appended claims, with the full scope of equivalents to which they may be entitled. It will be appreciated by those of ordinary skill in the art that any arrangement that is calculated to achieve the same purpose may be substituted for the specific embodiments shown. This application is intended to cover any adaptations or variations of embodiments of the present invention. It is to be understood that the above description is intended to be illustrative, and not restrictive, and that the phraseology or terminology employed herein is for the purpose of description and not of limitation. Combinations of the above embodiments and other embodiments will be apparent to those of skill in the art upon studying the above description. The scope of the present disclosed embodiments includes any other applications in which embodiments of the above structures and fabrication methods are used. The scope of the embodiments should be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled.

Claims

1. An adaptor for adapting a solar heater to a thermal storage device, said adaptor comprising: (a) a first port disposed on a first surface; (b) a second port disposed on a second surface; (c) a third port disposed on said second surface; and (d) a fourth port disposed on said first surface, wherein a fluid flow is configured to be directed from the solar heater through said first port before exiting through said second port into the thermal storage device, the fluid flow is configured to continue to be directed from the thermal storage device through said third port before entering through said third port and continuing to exit through said fourth port to return to the solar heater.
2. The adaptor of claim 1, further comprising a pump configured to direct the fluid flow from the solar heater to the thermal storage device and from the thermal storage device to the solar heater.
3. The adaptor of claim 2, further comprising a controller configured to control said pump.

4. The adaptor of claim 2, wherein an inlet of said pump is configured to be disposed at a level substantially the same as the level at which said third port is disposed and said third port is disposed at a level lower than the second port.
 5. The adaptor of claim 1, wherein at least one of said second port and said third port is configured for a quick connection or disconnection with the thermal storage device.
 6. The adaptor of claim 5, wherein at least one of said second port and said third port comprises a groove disposed about a periphery of said at least one of said second port and said third port and a seal disposed within said groove.
 7. The adaptor of claim 6, wherein said seal comprises an O-ring.
 8. The adaptor of claim 1, wherein the positioning of at least one of said second port and said third port relative to said second surface is configured to be adjustable.
 9. The adaptor of claim 1, further comprising an electrical receptacle for receiving an electrical connection from a solar electric generator.
 10. The adaptor of claim 9, further comprising an electric storage device configured to receive and store electrical charge from said solar electric generator.
 11. The adaptor of claim 10, further comprising an inverter configured to allow backfeeding of electricity from said electric storage device to an electrical grid through said inverter.
 12. The adaptor of claim 1, further comprising a hook configured for interfacing with a latch disposed on an enclosure of the thermal storage device such that a portion of the weight of the adaptor is supported on the enclosure while said second port and said third port are fluidly coupled with the thermal storage device.
 13. The adaptor of claim 1, further comprising a fifth port fluidly connected to said first port, wherein said fifth port is configured to be a bleed port.
 14. The adaptor of claim 1, further comprising a sixth port fluidly connected to said fourth port, wherein said fifth port and said sixth port are configured to be fluidly connected to a heat source.
 15. The adaptor of claim 1, wherein at least a portion of a fluid conductor connecting said first port and said second port is a flexible fluid conductor.
 16. The adaptor of claim 1, wherein at least a portion of a fluid conductor connecting said third port and said fourth port is a flexible fluid conductor.
 17. The adaptor of claim 1, wherein said first surface is disposed in a plane different from said second surface.
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